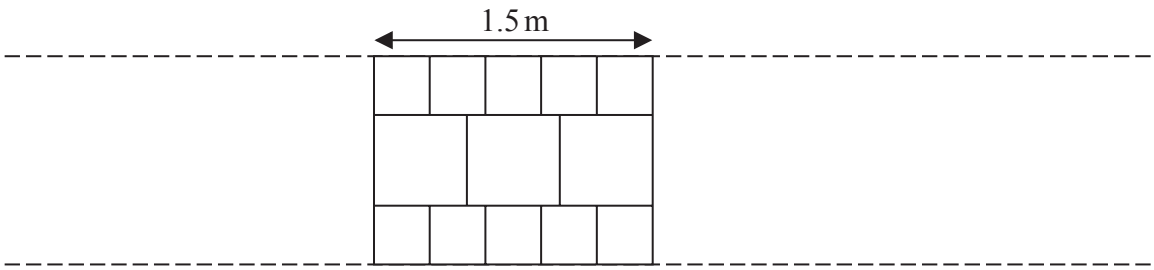
15 Jake is going to make a path from small paving stones and large paving stones.
The diagram shows Jake's design for the path.
The rest of the path is made using the same pattern of paving stones.



A small paving stone costs £2.30
A large paving stone costs £3.65

Jake needs to buy enough paving stones to make a path that is 6 metres long.

(a) How much will Jake have to pay for the paving stones he needs?

Each 1.5 m repeat has 10 small stones and 3 large stones.

m	Small	Large
1.5	10	3
×4	×4	×4
6	40	12

6 ÷ 1.5 = 4 repeats.

1 mark
Find 4 repeats

1 mark
40 small and 12 large

Cost = 40 × £2.30 + 12 × £3.65

= £92.00 + £43.80 = £135.80.

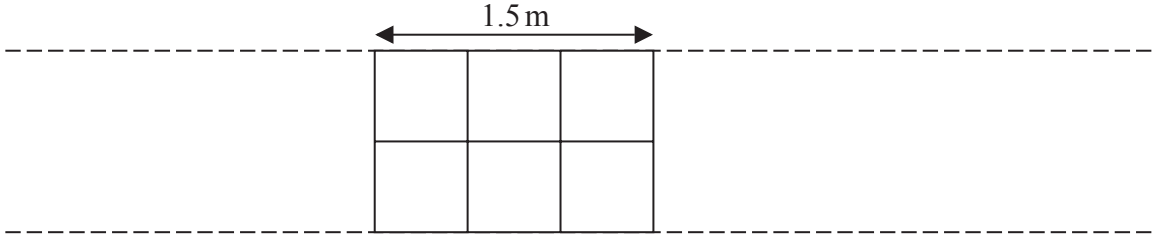
£.....

(4)

1 mark
Total both stone costs

1 mark
£135.80

Harry designs a different path that is also 6 metres long using the large paving stones.



Harry says that the cost of his path will be less than half of the cost of the path that Jake designed.

(b) Is Harry correct?
You must show how you get your answer.

Harry uses 6 large stones per 1.5 m repeat.

6 m needs 4 repeats: 6 × 4 = 24 large stones.

Harry's cost = 24 × £3.65 = £87.60.

Half Jake's cost = £135.80 ÷ 2 = £67.90.

No: £87.60 > £67.90, so it is more than half.

1 mark
Compare £87.60 with half of £135.80

1 mark
No; £87.60 > £67.90

(2)

(Total for Question 15 is 6 marks)